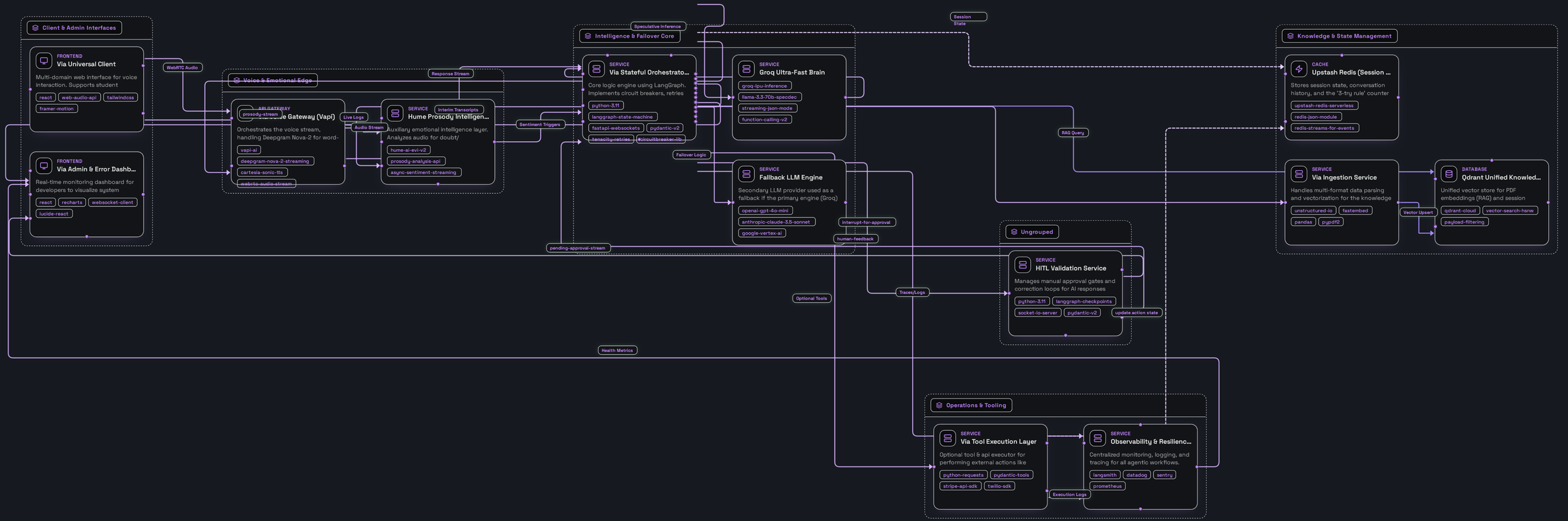
Client & Admin Interfaces
FRONTEND
Via Universal Client
Multi-domain web interface for voice interaction. Supports student
react
web-audio-api
tailwindcss
framer-motion
FRONTEND
Via Admin & Error Dashb...
Real-time monitoring dashboard for developers to visualize system
react
recharts
websocket-client
lucide-react
WebRTC Audio
Voice & Emotional Edge
API GATEWAY
prosody-stream
...e Gateway (Vapi)
Orchestrates the voice stream, handling Deepgram Nova-2 for word-
vapi-ai
deepgram-nova-2-streaming
cartesia-sonic-tts
webrtc-audio-stream
Live Logs
Audio Stream
SERVICE
Interim Transcripts
Hume Prosody Intelligen...
Auxiliary emotional intelligence layer. Analyzes audio for doubt/
hume-ai-evi-v2
prosody-analysis-api
async-sentiment-streaming
Response Stream
Sentiment Triggers
Speculative Inference
Intelligence & Failover Core
SERVICE
Via Stateful Orchestrato...
Core logic engine using LangGraph. Implements circuit breakers, retries
python-3.11
langgraph-state-machine
fastapi-websockets
pydantic-v2
tenacity-retries
pycircuitbreaker-lib
SERVICE
Groq Ultra-Fast Brain
groq-lpu-inference
llama-3.3-70b-specdec
streaming-json-mode
function-calling-v2
Failover Logic
SERVICE
Fallback LLM Engine
Secondary LLM provider used as a fallback if the primary engine (Groq)
openai-gpt-4o-mini
anthropic-claude-3.5-sonnet
google-vertex-ai
interrupt-for-approval
human-feedback
pending-approval-stream
Session State
Knowledge & State Management
CACHE
Upstash Redis (Session ...
Stores session state, conversation history, and the '3-try rule' counter
upstash-redis-serverless
redis-json-module
redis-streams-for-events
RAG Query
SERVICE
Via Ingestion Service
Handles multi-format data parsing and vectorization for the knowledge
unstructured-io
fastembed
pandas
pypdf2
Vector Upsert
DATABASE
Qdrant Unified Knowled...
Unified vector store for PDF embeddings (RAG) and session
qdrant-cloud
vector-search-hnsw
payload-filtering
Ungrouped
SERVICE
HITL Validation Service
Manages manual approval gates and correction loops for AI responses
python-3.11
langgraph-checkpoints
socket-io-server
pydantic-v2
update action state
Traces/Logs
Optional Tools
Health Metrics
Operations & Tooling
SERVICE
Via Tool Execution Layer
Optional tool & api executor for performing external actions like
python-requests
pydantic-tools
stripe-api-sdk
twilio-sdk
Execution Logs
SERVICE
Observability & Resilienc...
Centralized monitoring, logging, and tracing for all agentic workflows.
langsmith
datadog
sentry
prometheus